EYE-GAZE CONTROLLED WHEELCHAIR USING MACHINE LEARNING AND DEEP LEARNING APPROACH

PROBLEM STATEMENT

Eye gaze-based control for wheelchair movement is challenging, requires complex systems, and depends on costly assistive technologies. Accurate and real-time detection is lacking in accessible platforms. This creates difficulty in independent mobility and daily activities.

OBJECTIVE

Develop a real-time system for automated eye gaze detection, classification, and wheelchair control using computer vision.

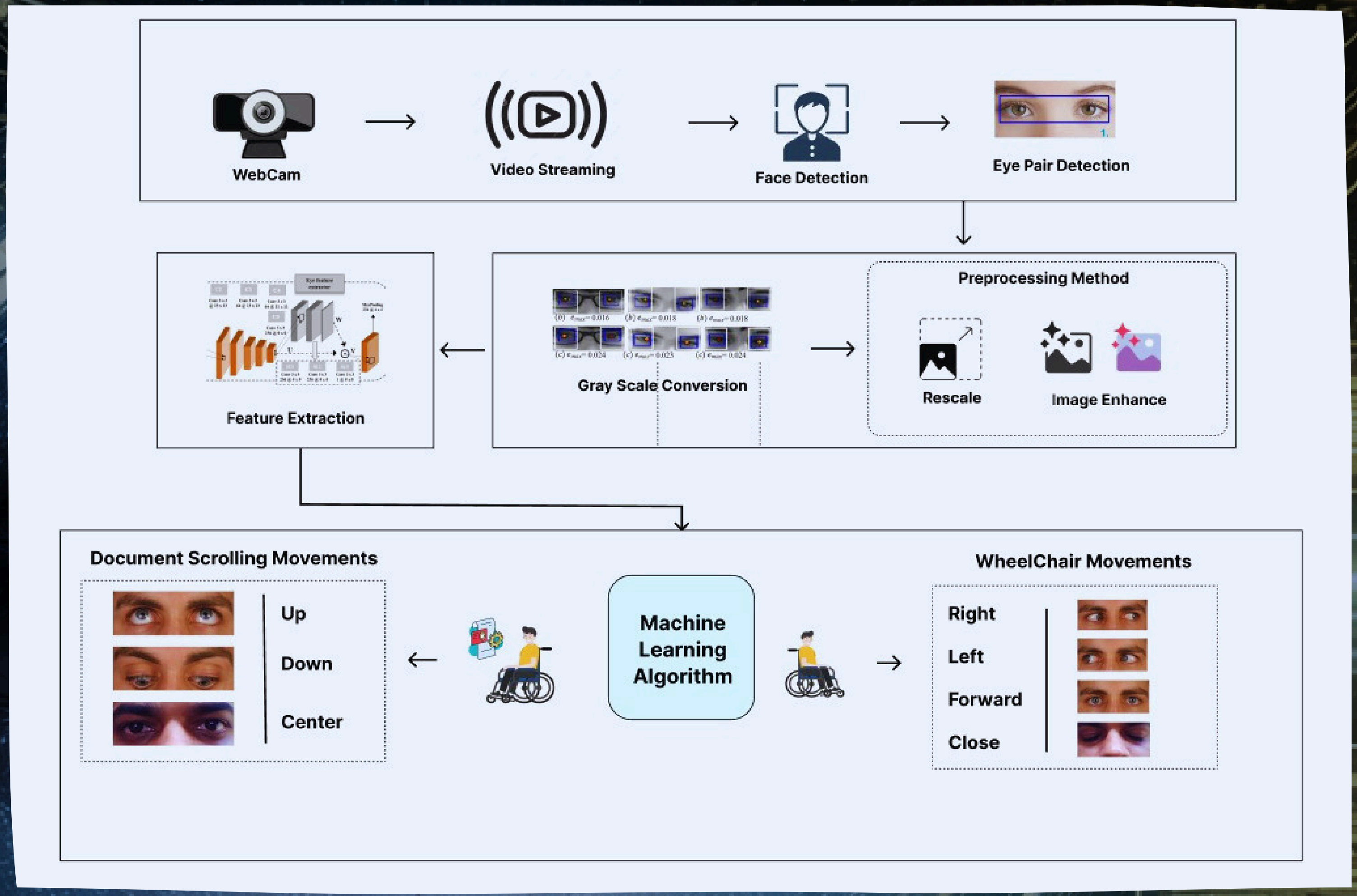
TECHNIQUE USED

Python-based system using MediaPipe/Haar Cascade for detection, OpenCV & NumPy for processing, and TensorFlow/Keras for gaze classification.

SDG	JUSTFICATION
SDG 3 - Good Health and Well-being	Enhances health through independent mobility.
SDG 4 - Quality Education	Supports accessibility using assistive technology.
SDG 9 - Industry, Innovation and Infrastructure	Promotes AI-based innovation.
SDG 10 - Reduced Inequalities	Reduces inequality for disabled users.

POs & PSOs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
Project Outcome	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

WORKING PRESENTATION



OUTCOME

Real-time eye gaze detection and accurate movement classification into left, right, forward, and stop. Smooth wheelchair control, low latency response, and visual simulation provide a reliable assistive solution. Not intended for medical use.

PSO1	Professinal skills
PSO2	Competency

MENTOR

Dr. A. Rajaprabu

STUDENTS

DHANARAJ R
GOWTHAM S
NITHISH S J

PO1	Engineering Knowledge
PO2	Problem analysis
PO3	Design/development of solutions
PO4	Conduct investigations of complex problems
PO5	Modern tool usage
PO6	The engineer & society
PO7	Environment & sustainability
PO8	Ethics
PO9	Individual & Team work
PO10	Communication
PO11	Project management & finance
PO12	Life-long learning